

Name: _____

Score: _____ / _____

Quiz 2

Part 1: Quiz 2

1

Consider the language $B = \{ (01)^n(10)^n \mid n \geq 0 \}$. For example, 010101101010 is in B, but 01010101010 is not in B. It is easy to check that every nonempty string w in B must satisfy all of the following properties:

- I. The number of 0s in w is the same as the number of 1s in w .
- II. w has exactly one instance of the substring 11, and it occurs in the middle of w .
- III. w does not have any instances of the substring 00.

Let p be the pumping length and let $s = (01)^p(10)^p$. s is in B and $|s| \geq p$, so, by the Pumping Lemma, s can be split into three pieces, $s = xyz$.

Choose why $xyyz$ is not in B. Select all the conditions I, II, and III above that are violated, **if y starts with a 1 and ends with a 0.**

(Example: If y starts with a 0 and ends with a 1, $xyyz$ has more symbols to the left of its occurrence of the substring 11 than to the right, violating condition II.)

A.

I

B.

II

C.

III

Answer Point Value: 1.0 points

Answer Key: B

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Let p be the pumping length and let $s = (01)^p(10)^p$. s is in B and $|s| \geq p$, so, by the Pumping Lemma, s can be split into three pieces, $s = xyz$.

Choose why $xyyz$ is not in B. Select all the conditions I, II, and III above that are violated, **if y starts and ends with a 1**.

(Example: If y starts with a 0 and ends with a 1, $xyyz$ has more symbols to the left of its occurrence of the substring 11 than to the right, violating condition II.)

A. I

B. II

C. III

Answer Point Value: 1.0 points

Answer Key: A,B

3

Consider the language $B = \{ (01)^n(10)^n \mid n \geq 0 \}$. For example, 010101101010 is in B, but 01010101010 is not in B. It is easy to check that every nonempty string w in B must satisfy all of the following properties:

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Choose why $xyyz$ is not in B. Select all the conditions I, II, and III above that are violated, **if y starts and ends with a 0**.

(Example: If y starts with a 0 and ends with a 1, $xyyz$ has more symbols to the left of its occurrence of the substring 11 than to the right, violating condition II.)

A.

I

B.

II

C.

III

Answer Point Value: 1.0 points

Answer Key: A,B,C

4

The Language of the following CFG, with starting variable R and alphabet $\{a,b\}$ is the complement of Palindrome.

$R \rightarrow XRX \mid S$

$S \rightarrow aTb \mid bTa$

$T \rightarrow XTX \mid X \mid \lambda$

$X \rightarrow a \mid b$

A.

True

B.

False

Answer Point Value: 1.0 points

Answer Key: A

5

For a given CFG **G**, we can always construct another CFG **H** such that, $L(\mathbf{G}) = L(\mathbf{H})$ and **H** is guaranteed not to be ambiguous.

True

False

Answer Point Value: 1.0 points

Answer Key: False

6

Let **G** be a CFG with starting variable **S** and let $\mathbf{A} = L(\mathbf{G})$. If we add a new rule $\mathbf{S} \rightarrow \mathbf{SS}$ to obtain a new grammar **H**, then $L(\mathbf{H}) = \mathbf{A}^*$.

True

False

Answer Point Value: 1.0 points

Answer Key: False

7

The language $\mathbf{PAL} = \{w \mid w = w^R\}$ can be used as an example to show CFL are not closed under complement.

True

False

Answer Point Value: 1.0 points

Answer Key: False

8

In our class definition, a Turing machine can transition from an accept state to reject state or vice-versa.

True

False

Answer Point Value: 1.0 points

Answer Key: False

9

A Turing Machine can never write the blank symbol on its tape.

True

False

Answer Point Value: 1.0 points

Answer Key: False

10

It is possible for the tape alphabet Γ of a Turing machine to be same as the input string alphabet Σ .

True

False

Answer Point Value: 1.0 points

Answer Key: False

11

There exist Turing machines that can be simulated by Deterministic Finite Automata.

True

False

Answer Point Value: 1.0 points

Answer Key: True

12

All languages must be at least one of the following:

i) Turing Recognizable

ii) Co-Turing Recognizable

iii) Decidable

True

False

Answer Point Value: 1.0 points

Answer Key: False

13

Given the encoding of a DFA D and a string w as input, it is possible to design a Turing Machine M that decides whether the string w is in the language of D .

True

False

Answer Point Value: 1.0 points

Answer Key: True

14

There exists a non-regular language L such that $\text{Prefix}(L)$ is a regular language.

True

False

Answer Point Value: 1.0 points

Answer Key: True

15

Decidable languages are not closed under the complement operation.

True

False

Answer Point Value: 1.0 points

Answer Key: False